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FDK™ Toolkit, Justice Domain Fairness Audit

External Validation Report: COMPAS Recidivism Dataset (Amended)

First Real-World Benchmark Validation Following Synthetic Diagnostic Remediation

Dataset: ProPublica COMPAS Recidivism Risk Score dataset (compas_dataset.csv), 7,214 defendants, Broward County, Florida, 2013–2014

Date: 23 July 2026 (amends the original report dated 21 July 2026)

Amendment notice: this report supersedes the version dated 21 July 2026. Bringing Justice's no-code UX in line with the five domains developed after it (see Findings 3–4) enabled a regression test against the same COMPAS data, which surfaced that the original validation's prediction-column mapping was incorrect (Finding 5). Section 4's fairness findings have been corrected accordingly; the qualitative conclusion is unchanged, but the disparity is substantially larger than originally reported.

Abstract

This report documents the first validation of the FDK™ Toolkit's Justice fairness-audit pipeline against a real, publicly available benchmark dataset, the ProPublica COMPAS recidivism risk-score dataset, following a comprehensive prior remediation program conducted entirely on synthetic diagnostic data. The initial validation run surfaced two genuine defects that had not manifested during synthetic testing: an all-or-nothing subgroup-size validation rule that rejected the entire 7,214-row dataset over a single small demographic group, and a missing-value handling gap in the feature-attribution-bias calculation. Both were fixed and verified at the time. This amendment documents three further findings from a subsequent review: Justice's no-code column-mapping UX, which did not exist when the domain was first validated, has now been built to match the five domains developed since (Finance, Health, Hiring, Business, Governance); the same statistical-fallback binary-detection gap found in every one of those domains was also present here and has been fixed; and, critically, exercising the new UX as a regression test revealed that the original validation's prediction column (is_recid) was not actually the COMPAS algorithm's output, but a near-duplicate ground-truth recidivism flag, meaning the original Section 4 compared the outcome against itself rather than against a genuine prediction. Corrected against the standard ProPublica decile_score-based binarization, the true disparity is substantially larger than originally reported: composite_bias_score rises from 0.087 to 0.343, and the headline selection-rate gap from 20.7 to 37.9 percentage points. The qualitative finding, a large, statistically robust racial disparity consistent with ProPublica's original 2016 investigation, is unchanged and, if anything, now more strongly corroborated.

1. Background and Purpose

Following a comprehensive internal remediation program, during which sixteen distinct defects were identified and resolved across all seven FDK™ Toolkit domain pipelines using purpose-built synthetic diagnostic datasets with hand-computed oracle values, the toolkit's Standard Operating Procedure called for a subsequent phase of validation against real or well-established public benchmark datasets to confirm that fixes verified on synthetic data generalize to genuine, messy, real-world input.

The Justice domain was selected for this first real-dataset validation using the ProPublica COMPAS dataset: a widely cited, extensively studied criminal-justice risk-assessment dataset that has itself been the subject of substantial public fairness scrutiny since ProPublica's original 2016 investigation. Its use here serves two purposes: confirming the FDK™ Toolkit's own correctness against real data, and reproducing a well-known fairness finding as an external sanity check on the toolkit's output.

This amendment follows from a separate review, conducted after Finance, Health, Hiring, Business, and Governance had each received a no-code confirmation-page and upload-page column-override UX, of whether Justice and Education (the two domains validated before that pattern existed) met the same standard. Justice's upload page was found to have no visible manual-override control at all (a hidden field only, producing no UI), and its confirmation page was static and read-only, exactly as Finance, Health, Hiring, Business, and Governance had each started before their own UX work.

2. Dataset Description

Property	Value
Total records	7,214 defendants
Screening date range	1 January 2013 – 31 December 2014
Group column used	race (6 categories)
Ground-truth column	two_year_recid (actual two-year reoffense, binary)
Prediction column (corrected)	predicted_high_risk, derived as decile_score >= 5 , the standard ProPublica binarization threshold, confirmed to align exactly with the dataset's own score_text categorization (Low vs. Medium/High)
Prediction column (original, incorrect)	is_recid , see Finding 5
Probability/risk-score column	Not mapped in this run (decile_score is an ordinal 1–10 field, not a continuous 0–1 probability; see Section 5)
Race group sizes	African-American: 3,696 · Caucasian: 2,454 · Hispanic: 637 · Other: 377 · Asian: 32 · Native American: 18

Two properties of this real dataset are directly relevant to the findings in Section 3: one demographic group (Native American, n=18) falls below the toolkit's minimum-subgroup-size threshold of 20 samples used elsewhere in this project; and one passthrough numeric column present in the raw file, violent_recid, is missing for all 7,214 records without exception.

3. Findings From Real-Dataset Validation

Finding 1, All-or-nothing subgroup rejection

The initial validation run rejected the dataset in its entirety, citing the Native American subgroup's 18 samples as below the 20-sample threshold, blocking analysis of the other 99.75% of the dataset entirely, including two groups with very strong statistical power.

Remediation: the hard rejection was replaced with a soft-exclusion policy. Groups below the 20-sample threshold are excluded from the analysis, with the exclusion recorded transparently, and the audit proceeds on the remaining adequately sized groups. This fix was later extended to Governance, which encountered the identical defect pattern against a genuinely real 17-person subgroup in the Folktables ACSPublicCoverage dataset.

Finding 2, Undefined result from a fully-missing feature column

Following the fix above, the pipeline returned feature_attribution_bias: NaN because the violent_recid column was present in the schema but entirely unpopulated, which invalidated the metric's averaging step across the whole dataset.

Remediation: the calculation now excludes only a column with zero non-missing values for a given group, rather than allowing one entirely missing column to invalidate the metric as a whole.

Finding 3, Statistical fallback accepted any two-valued column as binary

detect_justice_column_mappings' keyword-matched detection layer correctly required a candidate y_true/y_pred column's values to be a genuine subset of {0,1}. Its statistical fallback, however, checked only whether a column had exactly two unique values, with no verification of what those values actually were; the same defect pattern was found and fixed in every one of the five domains developed after Justice.

Remediation: the fallback now applies the same strict {0,1} check already used in the primary detection path.

Finding 4, No manual-override UX existed on either page

Justice's confirmation page displayed its auto-detected column mapping as static, read-only text, and its upload page carried only a hidden target_column field with no corresponding visible control, functionally identical to the starting state independently found and fixed in Health, Hiring, Business, and Governance.

Remediation: the same no-code dropdown-support functions (_describe_column, _candidate_columns, build_column_options) built for those four domains were constructed for Justice and wired into both pages: four real, filtered, pre-selected dropdowns on the confirmation page, and a header-row-parsed dropdown on the upload page.

Finding 5, Incorrect prediction-column mapping in the original validation

Exercising the newly built dropdown UX as a regression test against the same compas_dataset.csv used in the original validation surfaced that the original run's prediction column, is_recid, is not the COMPAS algorithm's output. It is a second, closely related ground-truth recidivism flag in the raw ProPublica data, distinct from but strongly overlapping with two_year_recid. The dropdown UX correctly identified is_recid as statistically valid (a genuine binary {0,1} column, not an identifier); the defect is semantic, not statistical, and no automated check in this project catches it. The original report's suspiciously perfect parity metrics (fnr_difference: 0.0, tpr_difference: 0.0, equal_opportunity_difference: 0.0, for_difference: 0.0) were themselves a symptom: two near-duplicate ground-truth columns will always agree almost perfectly, which looks statistically like a flawless, bias-free prediction rather than what it actually was, an outcome compared against itself.

Remediation: a genuine prediction column, predicted_high_risk, was derived using the standard ProPublica methodology (decile_score >= 5), confirmed to align exactly with the dataset's own score_text field. The audit was re-run with this corrected mapping; Section 4 below reflects the corrected result.

Lesson for future validation work: a technically valid column choice (binary, not an identifier, passes every automated check) can still be the wrong column semantically. Near-perfect parity across every error-rate metric is itself a signal worth checking manually, not just a good result to accept at face value, particularly when y_true and y_pred have similar-sounding names.

Verification of fixes

Check	Before fix	After fix
metrics_calculated	0 (rejected entirely)	35 / 35
feature_attribution_bias	NaN	0.4308
Statistical fallback accepts non-binary two-valued columns	Yes (undetected)	No, correctly rejected
Confirmation-page and upload-page column overrides	None (static display; hidden field with no UI)	Four real, filtered dropdowns; header-parsed upload dropdown
Prediction-column semantic validity	is_recid (near-duplicate ground truth, undetected)	predicted_high_risk (genuine decile_score-derived prediction)
composite_bias_score	0.087 (understated, based on incorrect mapping)	0.343 (correct mapping)

4. Fairness Findings on the COMPAS Dataset (Corrected)

With all five findings resolved and the prediction-column mapping corrected, the Justice pipeline produced a complete 35-metric audit across five race groups (African-American, Caucasian, Hispanic, Other, and Asian; Native American excluded per Finding 1). Corrected headline results:

Metric	Value	Interpretation
disparate_impact.ratio	0.356	Well below the conventional 0.8 four-fifths threshold; HIGH severity
statistical_parity_difference	0.379	A 37.9 percentage-point gap between the highest and lowest group (up from 20.7 points originally reported)
selection_rates_by_group (African-American)	0.588	58.8% flagged as high recidivism risk
selection_rates_by_group (Caucasian)	0.348	34.8% flagged as high recidivism risk
selection_rates_by_group (Other)	0.210	Lowest predicted-positive rate of the five analyzed groups (the lowest-rate group has changed from Asian to Other under the corrected mapping)
fpr_ratio	0.194	The most disadvantaged group's false-positive rate is roughly 5.2× the least-disadvantaged group's (up from 2.7× originally reported)
composite_bias_score	0.343	Up from 0.087 under the original, incorrect mapping
overall_assessment	HIGH_BIAS	Same label as originally reported, now reflecting the correct, substantially larger magnitude

These corrected findings are, if anything, more strongly consistent with ProPublica's original, widely cited 2016 analysis of this same dataset than the original (incorrect) mapping suggested. The FDK™ Toolkit's independent, automated audit, once pointed at a genuine prediction column, reproduces the same qualitative direction and a comparable magnitude of disparity to the original public investigation, which serves as a stronger external, real-world sanity check on the toolkit's correctness than the previous version of this report provided.

5. Scope Limitations of This Validation Run

- No probability/risk-score column was mapped for this run. COMPAS's native decile_score field is an ordinal 1–10 ranking, not a continuous 0–1 probability, and the toolkit's current y_prob detection requires the latter. A rescaled decile_score (e.g., (decile_score − 1) / 9) would allow calibration and AUC-based metrics to run in a future validation pass.
- temporal_fairness_score (0.111 in the original run) reflects COMPAS's genuine, uneven real-world screening-date distribution across a two-year window, rather than a defect; not independently investigated further as part of this amendment.
- This amendment corrects Section 4's numeric findings; anyone who previously relied on the original 21 July 2026 report's figures (composite_bias_score 0.087, a 20.7-point selection-rate gap) should treat those as superseded by the values in this version.
- The semantic-mapping defect in Finding 5 was found through manual review during a regression test, not by any automated check; no equivalent check currently exists in this project for other domains' validated datasets, so a similar undetected issue cannot be ruled out elsewhere without similar manual review.
- This validation covers the Justice domain only among the two domains (Justice, Education) validated before the shared no-code UX pattern existed. Education has not yet undergone the same review as part of this amendment.

6. Conclusion

The Justice domain fairness-audit pipeline has been successfully validated against a real, independently collected, publicly scrutinized benchmark dataset. Five genuine defects have now been identified, root-caused, fixed, and verified across the original validation and this amendment: two real-world data-handling gaps found during the original 2026-07-21 validation, a statistical-fallback detection gap and an absent no-code UX shared with four other domains, and, most significantly, an incorrect prediction-column mapping in the original validation itself, found only by exercising the newly-built UX as a regression test. The corrected fairness findings on COMPAS are more strongly consistent with established public research on this dataset than the original report's findings were, providing stronger external corroboration of the toolkit's correctness. The Justice domain is assessed as validated for real-world use under the corrected mapping, with the scope limitations in Section 5 noted for future work.

Appendix, Reference Materials

Dataset: compas_dataset.csv (ProPublica COMPAS Recidivism Risk Score dataset, Broward County, FL, 2013–2014)

Prior reports: FDK_Toolkit_Findings_Remediation_Report.docx; FDK_Toolkit_Production_Readiness_Report.docx; Justice_COMPAS_External_Validation_Report.docx (21 July 2026, superseded by this amendment)

Modified source files: fdk_justice_pipeline.py (calculate_all_metrics subgroup handling; _calculate_feature_attribution_bias missing-value handling; independent binary-validation check); fdk_justice.py (statistical-fallback fix; _describe_column, _candidate_columns, build_column_options; sanitization and override handling); auto_confirm_justice.html (static display replaced with filtered dropdowns); upload_justice.html (target-column dropdown added)

Validation reports referenced: justice_audit_report_20260721_155510.json (post subgroup-fix, pre-feature-attribution-fix); justice_audit_report_20260721_160350.json (original validation, is_recid mapping); justice_audit_report_20260723_164755.json (regression test, is_recid mapping reproduced); justice_audit_report_20260723_171304.json (corrected, predicted_high_risk mapping)